

# Grasp-Conditioned Planning and Learning: Supplementary Material

Anonymous Author(s)

TABLE I

PER-ROBOT ACQUISITION REWARDS IN THE SEPARATE PPO COMPARISON. RATES ARE MULTIPLIED BY  $\Delta t = 0.02$  s; EVENT REWARDS ARE ADDED ONCE. THE FROZEN FULL-SYSTEM ACTOR INSTEAD MINIMIZES IMITATION LOSS.

| Term                  | Rate / event value                                                      |
|-----------------------|-------------------------------------------------------------------------|
| Approach progress     | $6 \text{ clip}((d_{t-1} - d_t)/\Delta t, -0.1, 0.1)$                   |
| Proximity             | $2 \exp[-(d_t/0.10)^2]$                                                 |
| Alignment, original   | $\exp[-(\alpha_t/0.35)^2]$                                              |
| Alignment, gated      | $\exp[-(\alpha_t/0.35)^2 - (d_t/0.15)^2]$                               |
| Lift height           | $10H(z_t - z^*)$                                                        |
| Lift levelness        | $3 \exp[-\ \eta_t\ ^2/0.15^2]$                                          |
| Approach / lift costs | $-0.5 - 4(\beta_i/\beta_{\max})^2$                                      |
| Transition / failure  | +2 each at both-attached and lift-complete; -5 on physical termination. |

## I. SCOPE AND ORGANIZATION

This supplement provides reproduction details and secondary experiments for *Grasp-Conditioned Planning and Learning for Multi-Robot Transport*. The frozen planar mission study, pickup PPO comparison, spatial probes and native-contact studies use distinct protocols. Their outcomes are not pooled.

## II. REWARD SPECIFICATION

The full-mission actor minimizes the imitation objective in the weighted imitation objective in the main manuscript. This appendix specifies the separate pickup PPO and native-contact objectives; neither updates the frozen mission actor.

### A. Pickup objective and optimization

Two separate PPO branches each run 65,536 steps with an asymmetric 181-input critic [1], [2]. They use generalized advantages [3], a 0.3 KL anchor, value coefficient 0.5 and zero entropy coefficient. Table I defines their acquisition rewards; the variants differ only in distance gating of orientation reward. Transport rewards retain common command-tracking, posture, grasp and regularization terms, detailed in Table II. All evaluations use deterministic actor means.

For maximum hand-position/orientation errors  $d_t, \alpha_t$ , object roll/pitch  $\eta_t$  and base inclination  $\beta_i$ , the normalized height term is

$$\Phi(e) = (e^{-(e/0.08)^2} + 0.5e^{-(e/0.25)^2})/1.5, \quad (1)$$

$$H(e) = \max\left\{0, \frac{\Phi(e) - \Phi(e_0)}{1 - \Phi(e_0)}\right\},$$

where  $e = z - z^*$  and  $e_0 = z_{\text{rest}} - z^*$ . At zero initial error,  $H = \Phi$ . The inclination scale is  $\beta_{\max} = 2^\circ$ .

Both branches use one environment, horizon 128 and 512 updates; the learning rate is  $3 \times 10^{-5}$ , with three epochs, four minibatches, clip ratio 0.2,  $\gamma = 0.99$  and GAE  $\lambda = 0.95$ . Normalization is frozen and the critic receives 32 warmup updates. The network has ELU hidden layers 512–256–128. Mass is sampled from  $U[0.75, 3]$  kg, dimensions from  $U[0.9, 1.1]$ , base offsets from 0.08–0.65 m, headings within  $\pm 30^\circ$ , and activation delays within 0–2 s. COM perturbations are  $\pm(0.05, 0.12, 0.05)$  m and servo-gain factors 0.85–1.15. Automatic curricula are disabled.

Frozen pickup outcomes are 40/40 for the original reward and 39/40 for the distance-gated reward. Both stochastic training branches finish at 21/37 completed episodes. There is no demonstrated improvement from reward gating, and neither result establishes full-mission reliability.

### B. Transport objective

For  $K_\sigma(e) = \exp(-\|e\|^2/\sigma^2)$ , the command-normalized kernel is

$$b = K_\sigma(c), \quad k = K_\sigma(x - c),$$

$$T_\sigma(x, c) = \frac{k - b}{\max(1 - b, 0.1)} + \left[1 - \min\left(\frac{1 - b}{0.1}, 1\right)\right] k. \quad (2)$$

Table II specifies every transport term in the archived PPO environment. Approach/lift replace this sum with Table I; they do not add both objectives.

### C. Native-contact objective

The native-contact branch removes grasp welds and artificial attachment springs. Its per-robot objective combines progress, support and regularization with measured finger-contact costs:

$$r_{i,t} = \sum_{k \in \mathcal{K}_{\text{task}}} w_k f_{k,i,t} + \sum_{k \in \mathcal{K}_{\text{contact}}} w_k f_{k,i,t}. \quad (3)$$

Tables III and IV report guided-continuation (G) and the newer hand-target configuration (H) side by side; neither is retroactively assigned to earlier trials. Rates include  $\Delta t = 0.02$  s; progress, event, work and effort increments do not receive a second timestep factor. Both actors receive local observations; training aggregates their trajectories centrally.

For hands  $h \in \{L, R\}$ ,  $\langle x_h \rangle = \frac{1}{2} \sum_h x_h$ ,  $d_h$  is palm-to-target distance,  $c_h$  closure,  $q_h$  raw contact qualification,  $a_h$  retained contact status,  $f_h$  firmness and  $m_h$  friction margin.

TABLE II

ARCHIVED PPO TRANSPORT OBJECTIVE. EVERY ROW IS MULTIPLIED BY  $w_k \Delta t$ ; TERMINAL SEQUENCE FAILURE ADDS  $-5$  ONCE. THESE REWARDS DO NOT OPTIMIZE THE FROZEN IMITATION ACTOR.

| Term                  | Weight | Raw contribution $f_k$                           | Definition / gate                               |
|-----------------------|--------|--------------------------------------------------|-------------------------------------------------|
| Command task          | 10     | $e^{-\ (\xi - \xi^*) / (0.15, 0.15, 0.2)\ ^2}$   | Defines $P$ ; planar twist.                     |
| Linear tracking       | 3      | $T_{0.25}(v, v^*)$                               | No gate.                                        |
| Angular tracking      | 2      | $T_{0.35}(\omega, \omega^*)$                     | No gate.                                        |
| Height                | 2      | $H(z - z^*)$                                     | Multiplied by $P$ .                             |
| Level                 | 1      | $K_{0.15}(\eta)$                                 | Multiplied by $P$ .                             |
| Grasp                 | 1      | $\mathbf{1}[\text{engaged}_i]$                   | Multiplied by $P$ .                             |
| Base inclination      | 4      | $-(\max_j \beta_j / (\pi/90))^2$                 | No gate.                                        |
| Torso posture         | 0.5    | $K_{0.261799}(\psi_i)$                           | Multiplied by $P$ .                             |
| Arm symmetry          | 0.5    | $K_{0.15}(e_{\text{sym},i})$                     | Multiplied by $P$ ; metric hand-symmetry error. |
| Contact constellation | 1      | $-E_{c,i} / 0.05^2$                              | Mean squared palm-landmark error.               |
| Base constellation    | 1      | $-\mathbf{1}[\text{engaged}_i] E_{b,i} / 0.05^2$ | Mean squared base-landmark error.               |
| Residual magnitude    | 2      | $-\ a_{0:3}\ ^2 - 0.1\ a_3\ ^2$                  | Base and upper-body action costs.               |
| Object tilt           | 0.5    | $-\min(\ \eta\ _\infty / 0.15^2, 4)$             | No gate.                                        |
| Action change         | 0.005  | $-\ a_t - a_{t-1}\ ^2$                           | No gate.                                        |
| Drop                  | 50     | $-\mathbf{1}[z < z_{\text{rest}} - 0.12]$        | No gate.                                        |

TABLE III

COMPLETE NATIVE-CONTACT TASK OBJECTIVE. WEIGHTS G/H DENOTE GUIDED CONTINUATION / THE NEWER HAND-TARGET CONFIGURATION. ZERO DISABLES A TERM; TIMESTEP AND EVENT CONVENTIONS ARE EXPLICIT.

| Term                    | G    | H    | Raw contribution $f_k$                                    | Definition                              |
|-------------------------|------|------|-----------------------------------------------------------|-----------------------------------------|
| Grasp progress          | 20   | 20   | $\text{clip}(g_{t-1} - g_t, -.05, .05)$                   | Metric grasp-error reduction.           |
| Rotation progress       | 0    | 6    | $\text{clip}(\alpha_{t-1} - \alpha_t, -.02, .02)$         | Angular grasp-error reduction.          |
| Contact potential       | 5    | 5    | $\Phi_{i,t} - \Phi_{i,t-1}$                               | Distance, closure and retained contact. |
| Grasp loss              | 6    | 6    | $-N_{\text{release}}$                                     | Excludes injected self-opening.         |
| Proximity               | 1    | 1    | $\Delta t e^{-(g/.08)^2}$                                 | Dense approach reward.                  |
| Supported progress      | 100  | 100  | $G(\Delta s)$                                             | Path progress.                          |
| Goal progress           | 150  | 200  | $G(\ De_{t-1}\  - \ De_t\ )$                              | Translation and rotation.               |
| Supported lift progress | 0    | 200  | $G( e_{z,t-1}  -  e_{z,t} )$                              | Supported height-error reduction.       |
| Stable goal             | 2    | 2    | $\Delta t \mathbf{1}[\text{good}]$                        | Supported stable-goal condition.        |
| Unsupported travel      | 20   | 20   | $-\Delta \ell_{xy} \mathbf{1}[\text{unsupported}]$        | Horizontal object displacement.         |
| Anchor slip             | 2    | 2    | $-\Delta t \min(\delta, .1) \mathbf{1}[\text{engaged}_i]$ | Captured-frame grasp drift.             |
| Load                    | 0.2  | 0.2  | $-\Delta t (\ F_i\  / \max(1, 2F_{\text{cap}}))^2$        | Configured load normalization.          |
| Object orientation      | 4    | 4    | $-\Delta t \ e_R\ ^2$                                     | Nearby-reference rotation error.        |
| Torso posture           | 0.2  | 0.2  | $-\Delta t \psi_i^2$                                      | Local torso pitch.                      |
| Action change           | 0.01 | 0.01 | $-\Delta t \ a_t - a_{t-1}\ ^2$                           | Normalized actor outputs.               |
| Time                    | 2    | 2    | $-\Delta t$                                               | Per-second cost.                        |
| Success                 | 300  | 300  | $\mathbf{1}[\text{success}]$                              | Terminal bonus.                         |
| Failure                 | 500  | 500  | $-\mathbf{1}[\text{terminated}]$                          | Physical termination penalty.           |

The acquisition potential and support gate are

$$\Phi_i = \sum_h \left[ e^{-(d_h/0.04)^2} \text{clip}(c_h/c^*, 0, 1) + a_h \right],$$

$$G(v) = \begin{cases} v, & \text{supported at both interval endpoints,} \\ \min(v, 0), & \text{otherwise.} \end{cases} \quad (4)$$

The position-rotation error uses  $D = \text{diag}(1, 1, 1, \rho, \rho, \rho)$ ,  $\rho = 0.35$  m. Native contact shaping uses  $\sigma_d = 0.06$  m,  $\sigma_v = 0.05$  m/s and  $F_{\text{max}} = 40$  N; the separate load normalization uses  $F_{\text{cap}} = 60$  N. Finger work is  $\Delta W_i = \int \sum_j |\tau_j \dot{q}_j| dt$  and effort is  $\Delta E_i = \int \sum_j \tau_j^2 dt$ . The contact-frame cost uses four landmarks  $\{0, 0.04e_x, 0.04e_y, 0.04e_z\}$  per palm relative to the assigned or captured object frame. Recorded contact proxies are simulator diagnostics, not hardware-calibrated force-closure certificates.

For contact normals  $n_a$  on distinct finger bodies, the opposition score, normal-force total and force-weighted slip

are

$$o_h = \max_{a,b} \frac{1 - n_a^\top n_b}{2}, \quad F_{n,h} = \sum_a F_{n,a},$$

$$v_{\text{slip},h} = \frac{\sum_a F_{n,a} v_{\text{slip},a}}{\max(F_{n,h}, 10^{-12})}, \quad (5)$$

$$f_h = \min(F_{n,h}/12, 1) o_h,$$

$$m_h = \text{clip}(1 - \max_a \chi_a, 0, 1),$$

$$q_h = \mathbf{1}[o_h \geq 0.7, F_{n,h} \geq 2, v_{\text{slip},h} \leq 0.04],$$

where  $\chi_a$  is contact friction utilization. With no contact, all quality quantities are zero. First acquisition requires 0.08 s qualification; retained status tolerates at most 0.02 s unqualified contact, unless normal force falls below 0.2 N. These temporal filters affect observation and reward bookkeeping, not attachment forces.

The hand-target extension adds pose alignment, measured closure progress and relative held-hand motion, together with

TABLE IV

COMPLETE NATIVE-CONTACT CONTACT OBJECTIVE. WEIGHTS G/H DENOTE GUIDED CONTINUATION / THE NEWER HAND-TARGET CONFIGURATION. ZERO DISABLES A TERM; TIMESTEP AND EVENT CONVENTIONS ARE EXPLICIT.

| Term                 | G     | H     | Raw contribution $f_k$                                                     | Definition                                        |
|----------------------|-------|-------|----------------------------------------------------------------------------|---------------------------------------------------|
| First acquisition    | 8     | 8     | $\Delta N_i$                                                               | One event per hand's first qualified acquisition. |
| Distance             | 6     | 6     | $-\Delta t \langle \tanh(d_h / \sigma_d) \rangle$                          | Metric palm-target separation.                    |
| Separation speed     | 3     | 3     | $-\Delta t \langle \text{clip}(v_{\text{out},h} / \sigma_v, 0, 1) \rangle$ | Outward relative velocity.                        |
| Secure contact       | 4     | 0     | $-\Delta t \langle 1 - q_h \rangle$                                        | Qualified opposing contacts.                      |
| Firmness             | 1.5   | 1.5   | $-\Delta t \langle 1 - f_h \rangle$                                        | Contact-quality proxy.                            |
| Contact slip         | 12    | 12    | $-\Delta t \langle v_{\text{slip},h} \rangle$                              | Tangential slip speed.                            |
| Friction reserve     | 0.5   | 0.5   | $-\Delta t \langle 1 - f_h m_h \rangle$                                    | Firmness-weighted friction margin.                |
| Excess force         | 2     | 2     | $-\Delta t \langle \max(F_{n,h} / F_{\max} - 1, 0)^2 \rangle$              | Normal-force overage.                             |
| Finger work          | 0.02  | 0.02  | $-\Delta W_i$                                                              | Absolute actuator work increment.                 |
| Holding effort       | 0.002 | 0.002 | $-\Delta E_i$                                                              | Squared-torque integral.                          |
| Premature closure    | 0.5   | 0.5   | $-\Delta t \sum_h \mathbf{1}[d_{\text{nom},h} > .06] c_h^2$                | Closure away from the nominal target.             |
| Contact frame        | 100   | 100   | $-\Delta t \sum_h \text{mean}_k \ p_h + R_h u_k - p_h^* - R_h^* u_k\ ^2$   | Position and orientation via four landmarks.      |
| Pose alignment       | 0     | 3     | $-\Delta t \langle 1 - A_h \rangle$                                        | Assigned or captured palm frame.                  |
| Aligned closure      | 0     | 12    | $\Psi_t - \Psi_{t-1}$                                                      | Measured finger flexion, weighted by alignment.   |
| Held-hand adjustment | 0     | 1     | $-\Delta t \langle a_h J_h \rangle$                                        | Motion relative to the rigid object at the hand.  |

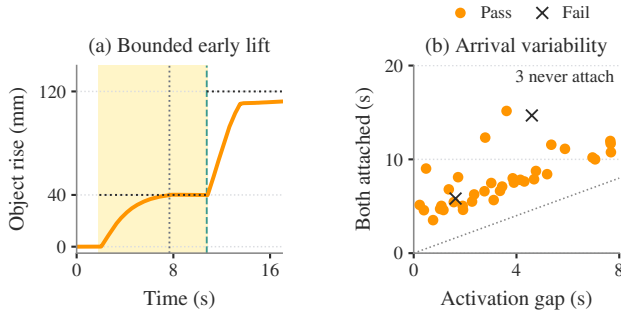


Fig. 1. Independent arrival. (a) Loading dock, seed 142013: measured object rise under a 7.68 s activation gap. Shading marks early lift; vertical dotted/dashed lines mark late activation/second attachment; horizontal lines mark height targets. (b) Both-attachment times for the 37 trials that acquire the object; color marks final mission outcome. The other three trials never attach both robots. The diagonal is  $t = \Delta$ .

supported lift progress:

$$A_h = e^{-(d_h/.03)^2 - (\alpha_h/.25)^2}, \quad (6)$$

$$\Psi_i = \langle A_h \text{clip}(\bar{c}_h/.6, 0, 1) \rangle.$$

Here  $\bar{c}_h$  is mean normalized MCP/PIP flexion of the non-thumb fingers, measured from joint encoders. Relative velocities subtract the object's rigid motion at the actual hand position. Tables III–IV specify both configurations independently.

$$J_h = \text{clip}\left(\frac{\|v_{\text{rel},h}\|^2}{.03^2} + \frac{\|\omega_{\text{rel},h}\|^2}{.2^2}, 0, 1\right). \quad (7)$$

### III. ASYNCHRONOUS ACQUISITION TRACE

#### IV. PLANAR IMITATION PROTOCOL

The mission actor is trained by imitation and DAgger [4]. A geometric teacher labels approach/lift, and a velocity actor labels transport. Three aggregation rounds collect 30 episodes each: teacher execution first, then student execution with teacher labels and no interventions. Stable-hold collection succeeds in 86/90 episodes. The final dataset contains 524,490 local observations, including 345,484 imported samples.

Phase-balanced weighted action MSE uses the settings in Table V and frozen normalization. The mission actor receives no subsequent PPO update.

Imported data comprise 227,902 samples from the earlier all-object actor and 117,582 from the independent-arrival actor, totaling 65.9% of the final dataset. In 22,205 early-lift rows, the height-reference feature was rebased from 120 to 40 mm; historical action labels, other features and observation noise were retained. These are reused development data, not an independent validation set. The initialization and dataset hashes are retained with the supplementary provenance record.

The transport teacher is the saved shape-conditioned velocity actor `velocity_v6_shapes`. In furnished scenes, collection commands zero transport velocity after lifting; navigation is supplied by RRT/MPC at test time. Thus successful collection is not evidence of learning complete doorway missions from demonstrations. The sampler balances the nonempty approach, lift and transport groups; the implementation minimizes

$$\mathcal{L} = \frac{1}{|B|} \sum_{j \in B} \frac{1}{21} \sum_{k=1}^{21} w_k (\pi_{\theta}(\bar{o}_j)_k - a_{jk}^*)^2. \quad (8)$$

Here  $\bar{o}$  uses frozen observation normalization. The bootstrap script's recorded defaults are batch size 512 and weights  $w = (4, 4, 2, 4, 0.5, 1, \dots, 1)$ ; equal phase allocation yields 510 samples when all three groups are present. The saved run metadata do not independently record every fitting CLI override, limiting exact reconstruction from metadata alone. Configuration and implementation details accompany the manuscript.

DAgger collection randomizes mass  $U[1, 2]$  kg, dimension factors  $U[0.95, 1.05]$ , rail friction  $U[0.9, 1.8]$ , base offsets 0.08–0.50 m, headings  $\pm 20^\circ$  and activation delays 0–8 s across ten scenes. Automatic stage/difficulty curricula are disabled; DAgger changes the executing teacher/student across rounds. Training ends at 35 s by truncation or earlier upon physical failure; successful DAgger collections also stop early. Figure 2 reports optimization and stochastic collection outcomes, distinct from the frozen evaluations.

TABLE V  
COMPACT LEARNING AND EXECUTION SETTINGS.

| Component            | Configuration                                     |
|----------------------|---------------------------------------------------|
| Actor input / output | 157 observations; 21 actions per robot.           |
| Network              | ELU MLP, 512–256–128 hidden units.                |
| Physics / policy     | 200 / 50 Hz; identical actor weights.             |
| Imitation            | Adam $3 \times 10^{-4}$ ; 3 fits, 40 epochs each. |

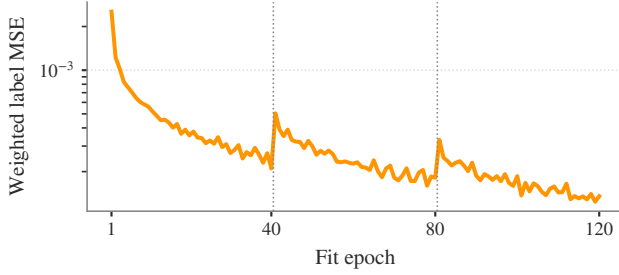


Fig. 2. Imitation fitting across three 40-epoch rounds; dotted lines mark dataset aggregation. Collection succeeds in 30/30 teacher episodes, then 28/30 and 28/30 student episodes. This is training evidence, not a matched comparison with the teacher at full-mission evaluation.

## V. SPATIAL REFERENCE AND PROBE PROVENANCE

The spatial imitation actor uses 54 collection episodes in three rounds, 60 fitting epochs per round, and planner-selected targets from  $\{0^\circ, 15^\circ, 45^\circ\}$ . Initialization transfers the upper/lower-grasp planar actor. After the first round, the configured teacher fraction is 0.25. The gallery-panel mass is fixed at 1.5 kg; activation is synchronized, without asynchronous early lift. Pose observations append local position/rotation error, desired and measured spatial twist, contact geometry and stance error. The actor still outputs 21 commands, with bounded arm/lift rates during spatial tracking. Pose references and phase information remain shared in training and execution. The  $90^\circ$  demonstration is geometric-teacher control, not this trained actor. These probes receive no PPO update; subsequent spatial PPO runs are separate development studies.

The illustrated teacher probes use upper/lower rail spacing 0.20 m; the learned probes use 0.16 m. A 0.15 m clearance lift takes 8 s after a 2 s post-lift hold. Angular interpolation takes 22.5 s for  $45^\circ$  and 45 s for  $90^\circ$ . The generic historical reference generator also permits  $180^\circ$  commands; the manuscript’s  $90^\circ$  fixed-grasp envelope is a declared scope restriction, not a retrospectively applied termination guard. Roll is relative to the initial lifted object frame. No large-roll probe establishes doorway passage or all-pose reachability.

The complete same-start learned test contains  $0^\circ$ ,  $15^\circ$  and  $45^\circ$  targets: two of three pass. The five website illustrations above are selected probes. The adaptive teacher index retains all 42 attempts (six passing), with changing geometry and controller settings; neither that count nor the selected successes estimates frozen-controller reliability. The failed  $180^\circ$  trial stops before  $90^\circ$  and therefore cannot identify  $90^\circ$  as an experimentally measured failure threshold.

TABLE VI  
SPATIAL PROBES. EACH ROW IS ONE NOMINAL TRIAL; CONFIGURATIONS DIFFER.

| Controller              | Goal        | Final roll    | Outcome          |
|-------------------------|-------------|---------------|------------------|
| Teacher                 | $45^\circ$  | $44.29^\circ$ | Pass             |
| Teacher                 | $90^\circ$  | $88.29^\circ$ | Pass             |
| Teacher (outside scope) | $180^\circ$ | $51.40^\circ$ | Tracking failure |
| Learned actor           | $15^\circ$  | $12.83^\circ$ | Pass             |
| Learned actor           | $45^\circ$  | $17.26^\circ$ | Self-penetration |

TABLE VII  
LATER FROZEN PLANAR COHORT: ALL TWELVE SCENES.

| Scene                | Completed / declared |
|----------------------|----------------------|
| Apartment            | 4/4                  |
| Bedroom              | 4/4                  |
| Open arena           | 4/4                  |
| Gallery panel        | 4/4                  |
| Living room          | 4/4                  |
| Living-room painting | 3/4                  |
| Loading dock         | 4/4                  |
| Office table         | 4/4                  |
| Dining room          | 4/4                  |
| Warehouse crate      | 4/4                  |
| Warehouse panel      | 3/4                  |
| Workshop beam        | 4/4                  |
| Overall              | 46/48                |

## VI. LATER PLANAR COHORT

The twelve-scene evaluation fixes the actor checkpoint and declares 48 starts before execution. It uses 1–2 kg loads,  $\pm 5\%$  dimensions, 0.08–0.50 m base offsets,  $\pm 20^\circ$  headings, 0.10 m object-position jitter and up to 8 s activation delay. The budget is 180 s and planned clearance is 20 mm. Upper/lower handles extend panel acquisition; supervised progression and idealized weld retention remain. Its different actor and source configuration prevent attribution of the difference from 35/40 to a single modification. The two acquisition timeouts occur in the living-room painting and warehouse panel scenes. Every declared trial remains in the denominator. This cohort has no large-roll task and does not replace the original formation/clearance analysis. Actor hash, source manifest and per-scene outcomes are retained in `spatial_evidence.json`.

## VII. LOCAL-HISTORY COORDINATION: IMPLEMENTED DEVELOPMENT TRIALS

A separate controller uses a shared causal transformer with private per-robot history, breakable finite attachments and local measurement ages. Its actor omits partner state, coordination phase and injected-fault flags. The spatial reference follows object progress; this differs from the timed spatial probes. A compact statement of the training/execution split is

$$a_{i,t} = \pi_\theta(o_{i,t-H+1:t}), \quad V_\psi(s_t^{\text{priv}}), \quad (9)$$

$$s_t^{\text{priv}} = [q_t, 0.1\dot{q}_t, b_{1,t}, b_{2,t}, f_t].$$

where  $q_t, \dot{q}_t$  contain both robots and the object,  $b_i$  are attachment indicators and  $f_t$  indicates fault injection. The centralized critic is used only in PPO training. Imitation

TABLE VIII

COMPLETED LOCAL-HISTORY DEVELOPMENT SUITES. COLUMNS ARE DISTINCT TASKS, NOT A POOLED SCORE.

| Variant | Transport | Wait | Early release | Motion |
|---------|-----------|------|---------------|--------|
| V44     | 12/24     | 4/8  | 6/8           | 14/14  |
| V45     | 12/24     | 7/8  | 7/8           | 12/14  |
| V46     | 12/24     | 3/8  | 7/8           | 13/14  |
| V47     | 0/24      | 0/8  | 0/8           | 0/14   |

centrally pools teacher-labeled robot samples; PPO pools rollouts and returns. Neither labels, partner histories, critic inputs nor optimizer state are supplied to the executing actors. Shared task references and simulated object estimates are still information inputs; no network budget is measured.

These four consecutive completed development variants are reported together rather than selecting the best score. V44 uses mixed-state imitation; V45 adds a current-observation head; V46 applies PPO; V47 varies release progress during collection. They are different training variants, not independent training seeds or a communication ablation.

The full suite contains 24 transport tasks and eight absent-partner waits. Transport comprises twelve nominal/delayed starts, eight later releases and four stale-pose cases. Later releases require four-hand support, object rise above 30 mm and path progress above 150 mm. Early-release and signed-motion suites have separate denominators. Safe waiting is not counted as transport completion. All variants reject payload/non-hand robot contact above 1 N; unsupported travel remains limited to 50 mm. V45 and V46 solve none of the eight later-release or four stale-pose cases. V47 fails all suites despite successful mixed-control collection, illustrating the gap between supervised fitting and closed-loop coordination. These results do not establish robust local recovery, broad scene transfer or learned 90° reorientation.

### VIII. FROZEN PLANAR FEEDBACK SEMANTICS

The frozen actor uses the legacy constraint feature: the first six equality solver multipliers for each palm weld are summed by row index, scaled by 0.01 and clipped to  $[-5, 5]$  before observation normalization. The first three rows constrain translation and the next three orientation. This concatenation is not transformed into a calibrated robot-frame spatial wrench; in particular, the rotational rows cannot be interpreted directly as wrist torques. It is a simulator-specific feature, not validated force sensing. Removing or replacing it requires evaluation with the appropriate observation distribution.

Contact gates test acquisition, but active welds can sustain tensile forces and moments without a frictional retention limit. Recorded actuator models do not, by themselves, validate hand force capacity. The study reports neither actuator saturation distributions nor a load-to-capacity ratio; the 1–2 kg objects test large geometry rather than heavy-payload capability. Frictional slip and hardware sensor equivalence remain untested. The separate local-history suites inject loss of support under finite attachments; those results do not validate the original welded-retention model.

TABLE IX

EVALUATION PROTOCOLS AND ORIGINAL DENOMINATORS.

| Test            | Domain / comparison                                                  | Pass  |
|-----------------|----------------------------------------------------------------------|-------|
| Frozen missions | 10 scenes; 1–2 kg; independent starts, $\Delta \leq 8$ s; 180 s.     | 35/40 |
| Original MPC    | 8 scenes; 1.5 kg; gated starts, $\Delta \leq 0.8$ s; 180 s.          | 36/40 |
| Aligned MPC     | Same actor and 40 seeds as original MPC; geometry and margin change. | 40/40 |

### IX. ADDITIONAL SPATIAL DIAGNOSTICS

### X. GEOMETRIC ILLUSTRATION

### XI. ADDITIONAL PLANAR DIAGNOSTICS

### XII. NATIVE-CONTACT ACQUISITION: SCOPE AND OUTCOMES

Native-contact experiments remove both grasp equalities and artificial hand–object springs. Load transmission instead depends on articulated finger contacts, finite actuator limits and randomized friction. Their objective terms appear in Tables III and IV. These experiments are separate from the welded mission evaluation and do not establish native-contact transport.

An actuator audit identified missing inheritance of hand motor defaults in the initial implementation. The corrected model sets finger/wrist position gains to 2 N m/rad and torque limits to  $\pm 1$  N m. A scripted four-hand acquisition, 6 cm lift and 8 s hold succeeds without attachment forces. Paired 500/1000 Hz simulations differ by at most 1.25 mm and 0.16° in object trajectory. This tests a short scripted trajectory, not learned robustness. Pre-correction failures remain in the archive and are excluded from corrected-model acceptance.

A paired 32,768-step PPO study uses identical initial weights, training seed and 20 evaluation seeds. The distance/outward-speed variant attains four-hand support in 18/20 trials but completes 0/20 lift-and-hold tasks: 17 fail on unintended payload–robot contact, two on self-contact and one on unsupported travel. The missing-contact-cost variant achieves neither acquisition nor completion (0/20); the baseline acquires in 2/20 and completes 0/20. These results distinguish contact acquisition from sustained task success; one training seed does not establish a general reward ranking.

Teacher initialization similarly fails to transfer reliably: collections with 10/12, 20/20 and 24/24 successful teacher trials each yield 0/20 fresh frozen-actor successes in their respective configurations. The corrected cross-scene screen completes 0/26 cases. Later corrective-label, architecture and reward experiments are development data and do not update these frozen cohorts. A subsequent calibrated near-grasp lift policy is reported in the expanded evidence section below. Full-mission native-contact reliability remains unestablished.

### XIII. PLATFORM AND PROSPECTIVE MOTOR INTERFACE

Each carrier has a holonomic three-module steering/drive base, a two-stage lift, a pitching torso, two 7-DoF arms and two hands with 16 finger DoFs and an additional wrist each (main manuscript architecture). The planar policy commands three base and eighteen upper-body coordinates;



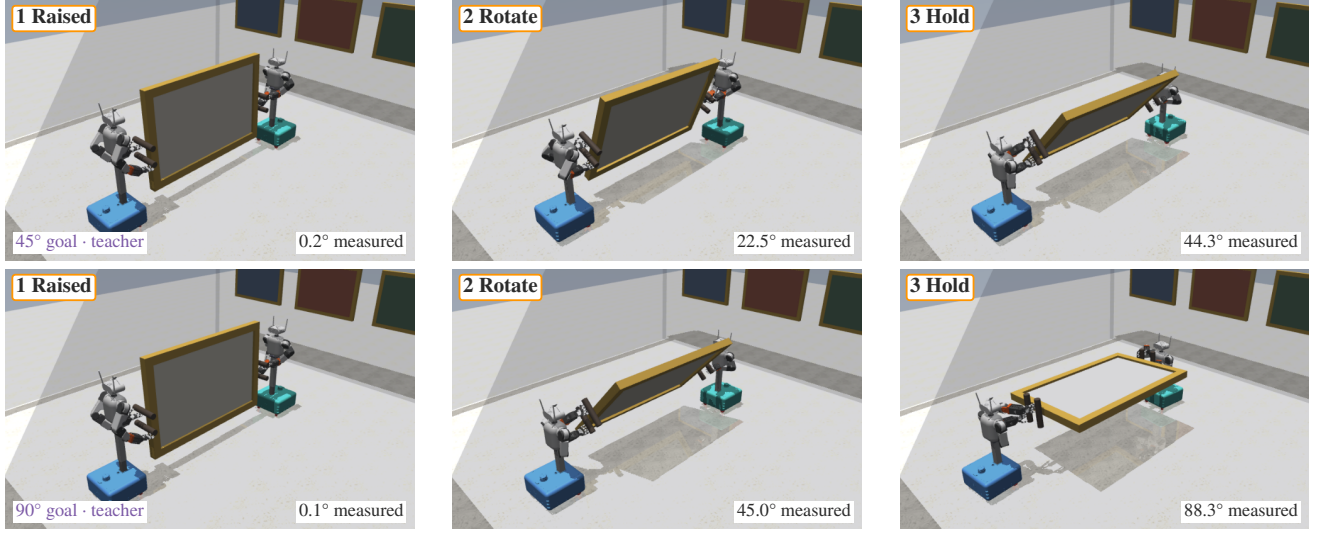


Fig. 3. Recorded fixed-grasp panel reorientation with the geometric teacher (seed 207000). Rows request 45° and 90° roll about the lifted object’s long axis. Columns show clearance lift, intermediate rotation and terminal holding; labels give measured angles. Robot A/B have blue/teal bases and native body materials. Original joint states are rendered without stepping dynamics. These are pose-only probes, not learned-policy or doorway-transport successes.

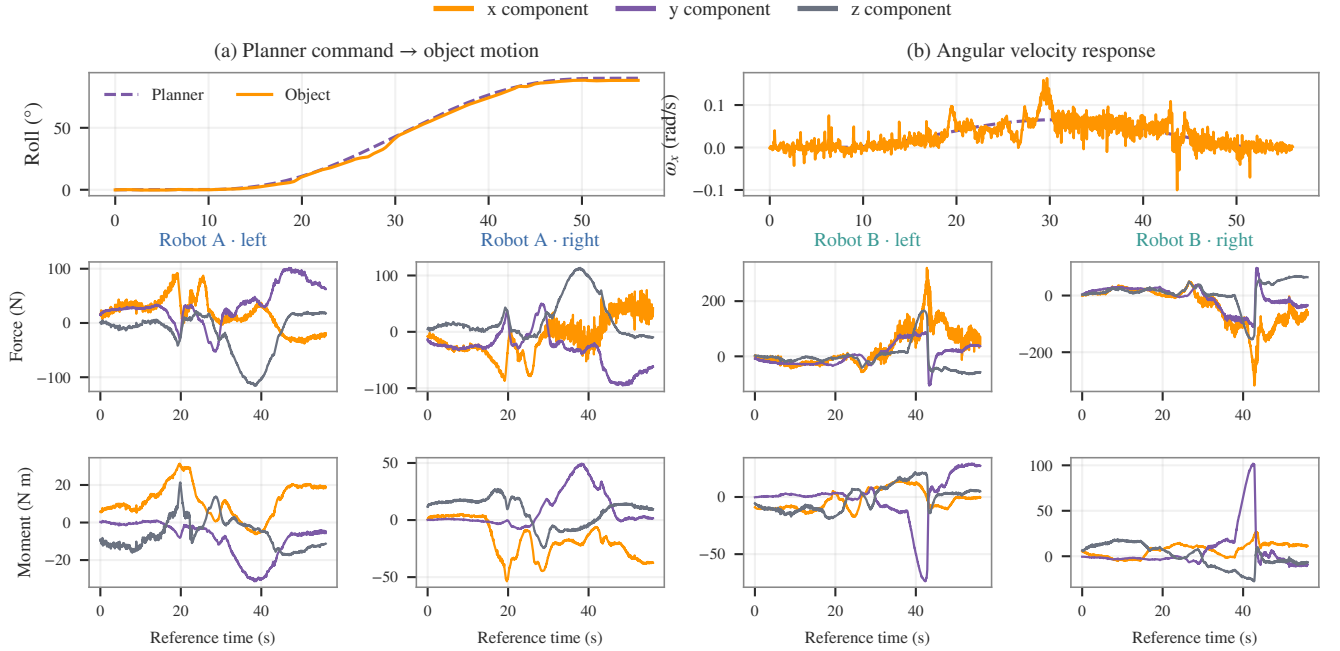


Fig. 4. Commanded object motion and resulting four-hand weld reactions in one instrumented 90° teacher probe. Dashed purple: pose/twist reference; orange: measured object response. Hand components use fixed lifted-object axes ( $x$ : orange,  $y$ : purple,  $z$ : gray); moments are palm-centered. Time starts at pose-schedule activation; ordinate scales vary by panel. These are simulated reactions, not commanded force targets or wrist-sensor data.

hand articulation uses stage-dependent closure rather than learned finger-level actions. The hardware interface uses Jetson Thor for local inference and CAN arm-drive commands ( $q^d, \dot{q}^d, K_p, K_d, \tau^{\text{ff}}$ ):

$$\tau^d = \text{clip}(K_p(q^d - q) + K_d(\dot{q}^d - \dot{q}) + \tau^{\text{ff}}, -\tau_{\text{max}}, \tau_{\text{max}}). \quad (10)$$

Embedded field-oriented control (FOC) regulates arm-motor current; MuJoCo uses its actuator model instead. The 4 kg hardware target load is distinct from the evaluated 1–2 kg simulation loads. Hardware system performance and that load

target are not validated by the reported experiments.

#### XIV. RECONSTRUCTION OF INDIVIDUAL HAND REACTIONS

For hand  $h$  of robot  $i$ , selected weld multipliers  $\lambda_{ih}$  give the payload wrench through

$$J_O^\top w_{ih}^{O,W} = S_O J_{ih}^\top \lambda_{ih},$$

$$w_{ih} = \begin{bmatrix} R_0^\top F_{ih}^W \\ R_0^\top [\tau_{ih}^{O,W} - (p_{ih}^W - p_O^W) \times F_{ih}^W] \end{bmatrix}. \quad (11)$$

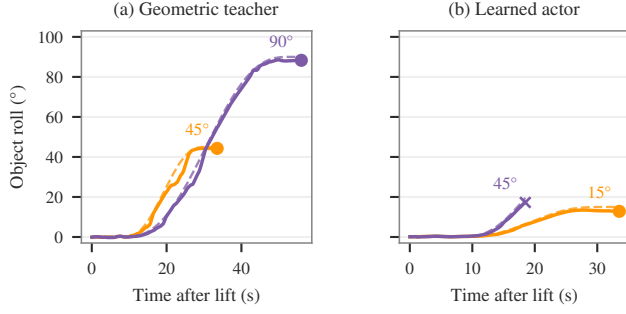


Fig. 5. Measured roll (solid) and evolving reference (dashed), relative to the lifted object frame. Circles mark successful completion; the cross marks self-penetration termination. Labels specify commanded endpoints, which are distinct from the evolving reference. Teacher and actor configurations differ.

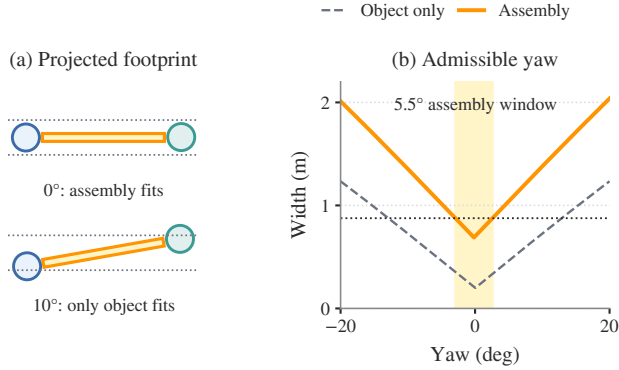


Fig. 6. Why the grasp changes passage feasibility. Analytic projections use the measured beam handoff geometry (seed 142024). (a) Dashed bounds show the available opening width after the 22 mm search margin on each side; the circles are base footprints. (b) The object-only yaw range is about 25.8°, versus 5.5° for the assembly (shaded); the horizontal dotted line is the door budget.

Here  $S_O$  selects the free payload’s six generalized coordinates,  $J_O$  maps their velocities to its world twist, and  $J_{ih}$  is the weld Jacobian.  $R_0$  fixes the lifted-object axes; moments act about the measured palm center  $p_{ih}$ . These solver reactions act on the payload and exclude separate finger-contact forces.

## XV. EXPANDED GEOMETRIC AND NATIVE-CONTACT EVIDENCE

### A. Grasp-space studies and matched search

The perimeter and end-face studies are separate enumerations. The former contains 2,110,211 opposed candidate layouts, 45,917 geometrically feasible layouts and 51 verified routes among 54 selected layouts. The latter checks 108 selected layouts with whole-body IK (95 feasible) and 54 selected routes (52 verified). These are discrete 50 mm screens and selected downstream checks, not coverage guarantees over continuous grasp assignments.

The new matched sampling study uses apartment-scene seeds 0–15, a 20,000-iteration budget, 0.35 m steering, goal bias 0.10, margin 0.02 m, 10 mm swept checking, explicit tree-join validation, no portal seeding and no smoothing. Gate probability alone changes, between 0.35 and zero. All returned

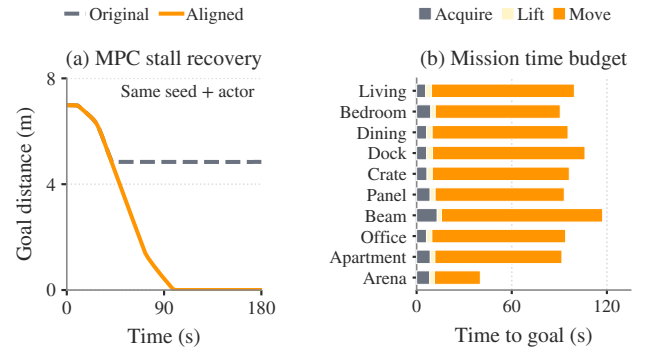


Fig. 7. Planning consistency and mission duration. (a) Same-seed, same-actor MPC comparison (114000): the original model stalls before the doorway; aligned geometry and margin permit progress to the goal. (b) Mean phase durations among successful frozen missions in each scene. Navigation dominates elapsed simulation time. The paired development case and frozen test use different actors and domains.

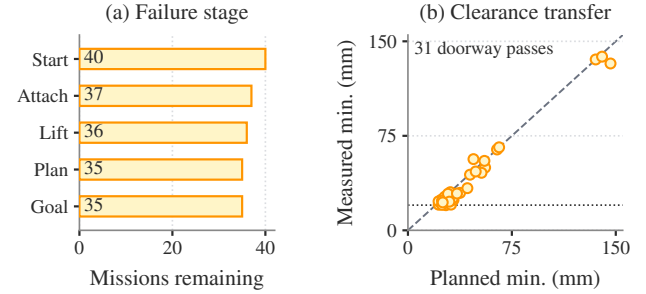


Fig. 8. (a) Mission counts at each stage, retaining all 40 starts. (b) Planned versus measured minimum clearance for 31 successful doorway missions; four arena passes have no finite clearance. Diagonal: equality; horizontal dotted line: 20 mm planning requirement.

routes are independently checked at 5 mm swept resolution. Verified counts are 16/16 and 5/16. Timing is recorded but not used to claim real-time capability; the machine was concurrently running other workloads. The legacy notebook’s eight-seed counts are not substituted for this study.

### B. Native checkpoint and task identity

The native lift checkpoint has SHA-256

e2ec616a001dbb03cea71ee68a434c090199  
eecef0e88e3246ad2c7d117b1ae7

and 81,920 environment steps (80 updates). Its frozen configuration contains an eight-frame, 2,816-input, two-layer 128–128 ELU actor, a 282-input critic and 23 actions per robot. PPO uses eight environments, horizon 128, four optimization epochs, four minibatches,  $\gamma = 0.995$ , GAE  $\lambda = 0.95$ , clip ratio 0.2, initial learning rate  $10^{-4}$ , entropy coefficient 0.0005 and frozen input normalization. Actor weights are not initialized from an imitation checkpoint. The two gripper output biases begin at +1 and remain trainable; initial log standard deviation is  $-0.8$ .

The base translation cap is 0.001 m/s and the base angular action scale is 0.001 rad/s; these are small nonzero authorities, not mechanical locks. Arm and torso position scales are 0.08 and 0.05 rad. The nominal payload is 0.3 kg, with calibrated palm offsets ( $-65, +70$ ) mm and no distant approach at

reset. Commanded height varies over 0.06–0.14 m. The independent hold-pose acceptance requires one continuous second with object position and orientation error below 0.04 m and 0.08 rad, every palm below 0.04 m and 0.175 rad error, every grasp qualified, and object speeds below 0.05 m/s and 0.2 rad/s. This differs from the older full-approach 8 s-hold evaluations; their rates are not pooled or treated as a matched improvement.

### C. Evaluation-domain correction

The 200-trial report uses seeds 7,500,000–7,500,199 and the fixed checkpoint above. All 200 trials satisfy the hold-pose criterion. Its original `difficulty=1.0` metadata predates the evaluator correction that writes this value into the environment’s randomization scale. Setting the training scheduler’s start value while disabling that scheduler did not change the environment. The manuscript therefore labels this cohort by the nominal randomization level and makes no full-domain claim from it. The original report is preserved byte-for-byte, with its interpretation in the evidence manifest. Subsequent full-domain evaluations and retraining are outside this frozen manuscript snapshot.

The evaluator also emits `time_limit` when a successful episode has no failure-cause string. This placeholder is not treated as evidence that all trials ran to their time budget. Reported completion times come from the measured first-success timestamp (median 5.07 s), and the 95% Wilson interval 98.1–100% is conditional on this task, checkpoint and seed domain. The training plot uses a rolling 20-completed-episode mean; no uncertainty band is inferred from a single training run.

### D. Reproducing the figures

The directory `data/ral/expanded` contains the selected original records, checkpoint configuration, source hashes, individual search trials and plot summaries. The original run was subsequently archived under `grasp_carry_L_lift_v1_weakDR`; its identity is the checksum above, not the reusable run-directory name. The figure script reads the frozen copies, so ongoing training cannot change the publication plots. The authored training configuration preserves base reward weights; the checkpoint snapshot can contain weights already rescaled by the curriculum. `make journal-figures` regenerates the three vector figures without robot simulation. The sampling benchmark is rerun separately through `scripts/benchmark_gate_sampling.py` in an M2 checkout.

### E. Frozen mission geometry, fitting and timing

The planar actor is trained by imitation and DAGger [4]. A geometric teacher labels acquisition/lift; a saved velocity actor labels transport. Three 30-episode aggregation rounds produce a 524,490-sample dataset, including 345,484 imported development samples. Training minimizes phase-balanced weighted action MSE across both robots; no subsequent PPO

TABLE X  
SCENE GEOMETRY AND FROZEN OUTCOMES: FOUR TRIALS PER SCENE.

| Scene     | Object   | $L \times D \times H$ (m)      | $S$ (%) | $T_L$ | $T_G$ |
|-----------|----------|--------------------------------|---------|-------|-------|
| Living    | Sofa     | $2.73 \times 1.12 \times 0.93$ | 75      | 9.0   | 98.3  |
| Bedroom   | Wardrobe | $2.10 \times 0.62 \times 0.95$ | 100     | 11.7  | 90.9  |
| Dining    | Table    | $2.40 \times 0.76 \times 0.75$ | 100     | 8.6   | 94.9  |
| Dock      | Case     | $2.80 \times 0.82 \times 0.80$ | 100     | 9.3   | 106.3 |
| Warehouse | Crate    | $2.20 \times 0.70 \times 0.65$ | 100     | 9.7   | 96.4  |
| Gallery   | Panel    | $1.90 \times 0.09 \times 1.25$ | 50      | 11.6  | 93.1  |
| Workshop  | Beam     | $3.20 \times 0.20 \times 0.20$ | 75      | 15.0  | 122.1 |
| Office    | Table    | $2.40 \times 1.10 \times 0.74$ | 75      | 9.1   | 93.5  |
| Apartment | Sofa     | $2.10 \times 0.86 \times 0.78$ | 100     | 10.7  | 91.8  |
| Arena     | Sofa     | $2.10 \times 0.90 \times 0.83$ | 100     | 11.3  | 40.3  |
| All (40)  |          | –                              | 87.5    | 10.6  | 95.4  |

Nominal dimensions; variation  $\pm 5\%$ .  $T_L/T_G$ : median seconds to completed lift/first goal among passes. Simulation time; planning compute is excluded.

update is applied. The supplement specifies data provenance, normalization and optimization settings.

Among successful missions, median lift completion and first goal arrival are 10.6 and 95.4 s; transport occupies a median 88.6% of simulated duration. Median search-plus-smoothing wall time is 26.45 s (11.41–70.70 s), excluding verification and retries. Simulation pauses during planning, so this time is excluded from mission duration. Complete MPC latency and deadline-miss distributions are unavailable; configured update rates do not establish real-time operation.

## XVI. REPRODUCTION RECORD

The accompanying `data/ral` records retain source hashes, imported-data provenance, reward definitions, actor/configuration identities and original outcomes. A review-response matrix distinguishes completed editorial and post-hoc analyses from missing controlled experiments. The spatial supplement freezes the site-linked probe sources and completed V44–V47 recovery suites; ongoing trials are excluded. The full-mission actor uses imitation; the reward table above is specific to this separate PPO study.

## XVII. STAGED ACQUISITION OF COMMANDED OBJECT POSE

The main manuscript’s native lift result is extended to commanded object motion by a four-stage curriculum in which each stage adds one degree of task freedom and is gated before the next is attempted: lift to a commanded height (L), translation of the held object (P1), rotation about all three axes (P2), and transport over metre-scale displacements (T). Every stage is graded with frozen schedulers on a held-out seed block under the full randomisation domain, and the acceptance conjunction requires: terminal object position and orientation, linear and angular stillness, four qualified contacts, all sustained for 1 s.

Stages L and P1 pass; stage P2 does not reach the 90% criterion (Table XI). A clause-resolved reading of the nine P2 failures attributes six to terminal position error (0.053–0.167 m), two to loss of qualified contact, and one to hold duration; no failure is attributable to orientation error alone.



TABLE XI

STAGED GATES UNDER THE STAGE ACCEPTANCE TOLERANCE (0.04 m, 0.08 rad). INTERVALS ARE WILSON AT 95 %.

| Stage | Added freedom         | Success       | 95 % CI   |
|-------|-----------------------|---------------|-----------|
| L     | commanded height      | 198/200       | 96.4–99.7 |
| P1    | translation           | 199/200       | 97.2–99.9 |
| P2    | rotation, three axes  | 51/60         | 73.9–91.9 |
| T     | metre-scale transport | not yet gated |           |

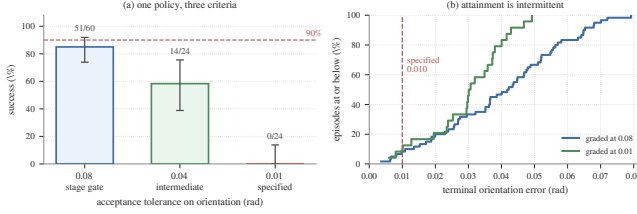


Fig. 9. Acceptance tolerance and attainment. (a) One frozen checkpoint under three orientation tolerances on identical held-out seeds; error bars are Wilson intervals and the dashed line is the 90 % criterion. (b) Empirical CDF of terminal orientation error; the distribution crosses the specified tolerance well below its median, locating the residual gap in consistency rather than attainable accuracy.

The failing objects are near-stationary at termination ( $0.0016\text{--}0.0090\text{ m s}^{-1}$ ), so they settle short of the commanded pose rather than fail to arrive within the episode.

#### A. Success rate is conditional on the acceptance tolerance

A single frozen checkpoint was graded under three orientation tolerances on the same held-out seeds (Fig. 9a). It passes 85.0 % at the stage tolerance of 0.08 rad, 58.3 % at 0.04 rad, and 0 % (0–13.8 at 95 %) at the 0.01 rad tolerance required by the task specification. The rate is therefore not a property of the controller in isolation, and a success rate reported without its acceptance tolerance is not interpretable. Terminal position is already inside the specified 0.04 m in 21 of 24 episodes, so the residual gap at the specified criterion is carried entirely by orientation.

#### B. The residual gap is one of consistency, not of attainable accuracy

The empirical distribution of terminal orientation error distinguishes two hypotheses that the mean cannot (Fig. 9b).

The minimum observed error is 0.0036 rad, between two and three times inside the specified tolerance, and the tenth percentile is 0.0108 rad, approximately at it; the median is 0.040 rad. The controller therefore attains the required accuracy in a minority of episodes and misses it on average. Two independent checks are consistent with this reading and inconsistent with an accuracy limit: the commanded arm increment under exploration is 0.001,21 rad per joint per step during carry, an order of magnitude finer than the tolerance; and the pregrasp inverse kinematics places all four palms within the 10 mm grasp criterion at commanded pitch up to  $90^\circ$ , so neither actuation resolution nor reachability binds at the relevant scale.

#### C. Scope

These are held-object results on a single payload under the stage acceptance tolerance, and they do not establish performance at the specified tolerance, on the multi-object requirement, or over the metre-scale transport stage, none of which is gated here. Commanded rotation in P2 spans  $\pm 0.07\text{ rad}$  about the pitch and yaw axes, an order of magnitude below the specification’s  $45^\circ$  minimum; the pitch envelope under load is not established by the reachability result quoted above, which is a kinematic bound and does not test moment transmission through the contact patch.

#### REFERENCES

- [1] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” *arXiv:1707.06347*, 2017. [Online]. Available: <https://arxiv.org/abs/1707.06347>
- [2] L. Pinto, M. Andrychowicz, P. Welinder, W. Zaremba, and P. Abbeel, “Asymmetric actor critic for image-based robot learning,” *arXiv:1710.06542*, 2017. [Online]. Available: <https://arxiv.org/abs/1710.06542>
- [3] J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel, “High-dimensional continuous control using generalized advantage estimation,” *arXiv:1506.02438*, 2015. [Online]. Available: <https://arxiv.org/abs/1506.02438>
- [4] S. Ross, G. Gordon, and D. Bagnell, “A reduction of imitation learning and structured prediction to no-regret online learning,” in *Proc. Int. Conf. Artif. Intell. Stat. (AISTATS)*, ser. Proceedings of Machine Learning Research, vol. 15, 2011, pp. 627–635. [Online]. Available: <https://proceedings.mlr.press/v15/ross11a.html>